Unit – 3
Mixed Reality

Introduction to Mixed Reality

- **Mixed Reality (MR)** is an advanced technology that blends the **real world and virtual world**, allowing physical and virtual objects to **co-exist and interact in real time**.

- It lies between **Augmented Reality (AR)** and **Virtual Reality (VR)** in the Reality–Virtuality Continuum

- Proposed by Paul Milgram.

Reality–Virtuality Continuum

- Real Environment (RE) → Completely real world
- Augmented Reality (AR) → Real world + digital overlays
- Mixed Reality (MR) → Real & virtual objects interact
- Augmented Virtuality (AV) → Virtual world + some real elements
- Virtual Reality (VR) → Fully immersive virtual world

Definition of Mixed Reality

- Mixed Reality is a technology that:
  - Combines physical and digital content
  - Embeds virtual objects to the real world
  - Enables real-time interaction between real and virtual objects
  - Allows users to manipulate both environments naturally
- MR understands the **environment spatially**, making virtual objects behave as if they are part of the real world.

Key Features of Mixed Reality

- **1. Spatial Awareness**
  - MR devices map the physical environment.
  - Objects can be placed on real tables, walls, floors.
- **2. Real-Time Interaction**
  - Users can move, resize, rotate virtual objects.
  - Digital objects respond to physical surfaces.

Key Features of Mixed Reality

- **3. Environmental Understanding**
  - Recognizes walls, furniture, depth, and surfaces.
  - Avoids placing objects through walls.
- **4. Persistence**
  - Virtual objects remain in the same location even after restarting the device.
- **5. Natural User Interface**
  - Interaction through gestures, voice, eye tracking, and controllers.

Components of Mixed Reality System

- **A) Hardware Components**
- **Display Devices**
  - Head-Mounted Displays (HMDs)
  - Smart glasses
  - MR headsets like Microsoft HoloLens
  - Magic Leap 1
- **Sensors**
  - Cameras
  - Depth sensors
  - Accelerometers
  - Gyroscopes
  - LiDAR (in advanced devices)

Components of Mixed Reality System

- A) **Hardware Components**
  - **Processing Unit**
    - CPU, GPU
    - Dedicated AI processors
  - **Input Devices**
    - Hand tracking
    - Controllers
    - Voice commands
    - Eye tracking

Components of Mixed Reality System

- B) **Software Components**

  - Spatial Mapping Software
  - Rendering Engine
  - Tracking Algorithms
  - Interaction Engine
  - Development Platforms
    - Unity
    - Unreal Engine
    - Microsoft Mixed Reality Toolkit

How Mixed Reality Works

- **Step-by-Step Process:**
- **Environment Scanning**
  - Cameras & sensors scan surroundings.
- **Spatial Mapping**
  - System creates 3D mesh of environment.
- **Tracking**
  - Tracks user's head, hands, and movement.
- **Content Placement**
  - Virtual objects are anchored in real space.
- **Rendering**
  - 3D objects are rendered with correct lighting & perspective.
- **Interaction**
  - User interacts using gestures or voice.

Applications of Mixed Reality

- **1. Education**
  - Virtual lab experiments
  - Interactive 3D models
- **2. Healthcare**
  - Surgical training
  - Anatomy visualization
- **3. Engineering & Manufacturing**
  - Product design
  - Maintenance assistance

Applications of Mixed Reality

- **4. Architecture**
  - Virtual building walkthroughs
- **5. Gaming**
  - Interactive real-world gaming
- **6. Remote Collaboration**
  - Holographic meetings

Advantages of Mixed Reality

- Enhances learning and understanding
- Improves productivity
- Reduces training cost
- Safe simulation environment
- Better visualization

Limitations of Mixed Reality

- Expensive hardware
- Limited battery life
- Processing power requirement
- Privacy concerns
- Limited field of view in current devices

Thank You…!!!